Q.1 Describe all three data set in terms of following : how many features, their categories (num,text,categorical, nominal,ordinal etc), number of tuples, missing values.

1. Iris Dataset

Features: 5

4 Numeric (Sepal Length, Sepal Width, Petal Length, Petal Width)

1 Categorical Nominal (Species)

Number of Tuples: 150

Missing Values: None

2. Air Quality Dataset

Features: 15

Numeric (gas sensor measurements, temperature, humidity)

Text/Temporal (Date, Time)

Number of Tuples: 9358

Missing Values: Yes (represented as -200)

3. Wine Quality Dataset

Features: 12

11 Numeric (acidity, pH, alcohol, etc.)

1 Categorical Ordinal (Quality)

Number of Tuples:

1599 (Red wine)

4898 (White wine)

Missing Values: None

**Q.2 In IRIS data set which non graphical EDA is performed ? **

1. Descriptive Statistics

Descriptive statistics summarize the main characteristics of the numerical features.

For the four numerical attributes (sepal length, sepal width, petal length, petal width) we calculate:

Mean – average value of each feature

Median – middle value of the data

Minimum and Maximum – smallest and largest values

Standard Deviation – variation or spread of data

Variance – measure of dispersion

These statistics help understand the central tendency and spread of the dataset.

2. Frequency Distribution of Categorical Variable

The Species attribute is a nominal categorical variable. Frequency analysis is performed to count the number of records in each class.

Example:

Species Count Setosa 50 Versicolor 50 Virginica 50

This shows the dataset is balanced.

3. Correlation Analysis

Correlation analysis measures the relationship between numerical variables.

Example findings:

Petal length and petal width have strong positive correlation.

Sepal width has weaker correlation with other variables.

Correlation values usually range from -1 to +1.

4. Summary Statistics by Class

Statistics are calculated separately for each species.

Example analysis:

Average petal length for Setosa

Average sepal width for Versicolor

Average petal width for Virginica

This helps identify differences between species.

Q.3 n Air Quality data set which graph is used to perform multivariate analysis?

For air quality data sets, which typically involve multiple pollutants (CO) and meteorological factors (temperature, wind speed, humidity), several graphs are used to perform multivariate analysis to understand relationships, correlations, and patterns. The most effective graphs for this purpose are:

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Heatmap (Correlation Matrix): A heatmap is highly effective for determ
and, or negative correlation with wind speed).
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Scatter Plot Matrix (Pairplot): This graph shows all pairs of X-Y plot

Parallel Coordinates Plot: This visualizes high-dimensional data, mapp

Biplot (PCA): Principal Component Analysis (PCA) biplots combine score
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Q.4 In Wine quality data set can you do non graphical EDA ? Explain your answer.

How to Perform Non-Graphical EDA on Wine Quality Data Using Python's pandas library, you can perform the following non-graphical techniques to understand the dataset (which includes features like fixed acidity, volatile acidity, residual sugar, pH, and alcohol):

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Viewing Data Structure (.head(), .tail(), .sample()): Inspecting the f

Checking Data Info (.info()): This reveals the number of entries, colu
Checking for Missing Values (.isnull().sum()): This identifies missing

Descriptive Statistics (.describe()): This provides count, mean, stand

Correlation Matrix (.corr()): By calculating pearson correlation coeff

Value Counts (.value_counts()): This is used to analyze the distributi
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Example Non-Graphical Findings on Wine Data Based on the Wine Quality dataset, non-graphical techniques usually reveal:

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No missing values: Most versions of this dataset are complete.

Highly skewed features: residual sugar and chlorides often have large

Core Correlations: A .corr() analysis typically reveals that alcohol h
```